Engineering - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Use Laplace's equation to find the capacitance of a parallel platecapacitor.
    (a) C= (ϵ^2S/ d)
    (b) C= (/ d^2)
    (c) C= (/ d)
    (d) C= (ϵ/ dS)
2. Speed of data transmission in 4-G network of telecom is
    (a) 10 gbps - 100 gbps.
    (b) 50 mbps - 500 mbps.
    (c) 2 mbps - 1 gbps.
    (d) 100 mbps - 1 gbps.
    (e) 386 kbps - 2 mbps.
3. If a Hexadecimal number needs to convert to binary. For each hexadecimal digit, there will be how many bits
    (a) 1
    (b) 2
    (c) 4
    (d) 8
4. What is the name of the fluorescent material that gives red colour fluorescence?
    (a) Zinc silicate.
    (b) Calcium sulphide.
    (c) Zinc oxide.
    (d) Zinc sulphide.
    (e) Calcium silicate.
5. The feedback factor of a Wien bridge oscillator using Op-Amp is
    (a) 3/2
    (b) 1/4
    (c) 0
    (d) 1/5
    (e) 1/3

True / False (10 marks)

Answer True or False

6. True or False: The probability of obtaining 2 heads and 1 tail when tossing a penny 3 times is 3/4.

7. True or False: The correct answer to ’Find the input impedance of a 50 ohm line terminated in + j50 ohms, for a line length such that = π radian.’ is ’100 ohms’.

8. True or False: The Laplace inverse of F(s) = [(5s^2 - 15s + 7) / {(s + 1)(s - 2)^3}] is e^-t + 2e^2t - 3te^2t + (t^3 / 6)e^2t.

9. True or False: The normal acceleration for the fluid particle is given by a_n = [(2ca) / (a^2 + b^2)].

10. True or False: The probability of drawing one red ball from the box is 5/14 and the probability of drawing one black ball is 4/14.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the unit vector direction in which a positive test charge should be moved from the point (1, 3, -1) in the electric field E = -xa_x + ya_y + 3a_z to achieve both maximum opposing and aiding forces.

12. Calculate the efficiency of a 7-hp motor that consumes 6.3 kilowatts at full load. Discuss the implications of this efficiency in practical engineering applications.

End of Paper